

HARISH KASU

AI/ML Engineer

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SUMMARY

- AI/ML Engineer with 5 years of experience designing and deploying scalable ML, NLP, and Generative AI systems across cloud, edge, and enterprise environments.
- Expertise in developing and fine-tuning Large Language Models using LoRA, QLoRA, RLHF, and Constitutional AI for domain specific and safe AI applications.
- Skilled in building Retrieval-Augmented Generation pipelines and multi-modal AI systems integrating text, vision, and tabular data for intelligent enterprise workflows.
- Proficient in MLOps, deploying production-grade models using MLflow, Airflow, Kubeflow, Docker, and Kubernetes with automated CI/CD and multi-cloud orchestration.
- Experienced in GPU optimization using TensorRT, DeepSpeed, and Triton Inference Server, enhancing model throughput, latency, and inference efficiency.
- Committed to responsible AI through model explainability, fairness, and transparency using SHAP, LIME, Captum, and bias mitigation frameworks.

PROFESSIONAL EXPERIENCE

NVIDIA
AI/ML ENGINEER

May 2024 – Present
CA, USA

- Architected and deployed retrieval-augmented generation (RAG) pipelines using LangChain, FAISS, and NVIDIA NeMo, powering domain specific generative AI applications for enterprise clients with a 42% improvement in retrieval precision.
- Fine-tuned foundation models (Llama 3, Mixtral, Claude) using LoRA/QLoRA on multi-GPU clusters with NVIDIA A100s, achieving 30% faster training throughput and optimized VRAM utilization with DeepSpeed and bitsandbytes.
- Developed multi-modal generative AI systems combining text, image, and tabular modalities using NVIDIA NeMo and PyTorch Lightning, enabling rich contextual understanding for enterprise document intelligence workflows.
- Led design of custom CUDA-optimized inference pipelines for LLMs and diffusion models, integrating TensorRT LLM and Triton Inference Server, reducing response latency by 45%.
- Implemented LLM evaluation and benchmarking framework leveraging LangSmith, Weights & Biases, and MLflow, standardizing model assessment across accuracy, safety, and latency metrics.
- Collaborated cross-functionally with the NeMo and DGX platform teams to integrate quantization (INT8/FP16) optimization and speculative decoding into production inference, improving cost efficiency and scalability.
- Built end-to-end MLOps pipelines with Kubeflow, Airflow, and GitHub Actions, automating data versioning, model training, validation, and deployment across multi-cloud environments.
- Developed vector search APIs for internal knowledge retrieval using FAISS and Milvus, integrated with LangGraph-based autonomous agents, enabling intelligent document and codebase discovery for R&D teams.
- Implemented continuous monitoring and drift detection for LLM services using Prometheus, Grafana, and NVIDIA Morpheus, ensuring SLA compliance and proactive retraining triggers.
- Partnered with Responsible AI team to apply bias mitigation and red-teaming pipelines, conducting safety alignment of deployed LLMs with Constitutional AI principles.
- Mentored junior engineers on prompt engineering, fine-tuning workflows, and GPU optimization techniques, contributing to internal NVIDIA AI Guild knowledge base.
- Delivered proof-of-concept generative agents using CrewAI and LangGraph for automated testing, documentation, and model evaluation tasks, increasing internal productivity by 28%.

Microsoft Corporation Azure
AI/ML ENGINEER

April 2020 – July 2023
Hyderabad, India

- Designed and implemented end-to-end ML pipelines on Azure Machine Learning, automating data ingestion, preprocessing, model training, and deployment by reducing model lifecycle time by 35%.
- Collaborated with the Azure AI team to develop and optimize NLP and CV models leveraging PyTorch, TensorFlow, and Hugging Face Transformers for enterprise-scale applications.
- Supported Generative AI feature integration into Azure Cognitive Services by building and testing custom RAG workflows using Azure OpenAI Service, LangChain, and Azure Cognitive Search.
- Built and deployed real-time inference APIs using FastAPI and Azure Kubernetes Service (AKS), improving latency by 40% and enabling scalable model serving for production workloads.
- Developed data preprocessing and feature engineering pipelines using Azure Data Factory, Databricks, and Pandas, ensuring high-quality, consistent inputs for ML model training.
- Assisted in fine-tuning foundation models (GPT-3, Llama 2, and T5) using LoRA/QLoRA and RLHF techniques, enhancing model accuracy for domain-specific tasks like summarization and Q&A.
- Implemented monitoring and drift detection systems using Azure ML, MLflow, and Application Insights, enabling proactive model retraining and improving long-term model stability.
- Collaborated cross-functionally with data scientists and product teams to evaluate model performance using metrics such as F1-score, precision, recall, and AUC, delivering a 15% uplift in prediction accuracy.
- Built automated CI/CD pipelines for ML workflows using GitHub Actions and Azure DevOps, enabling seamless deployment of updated models and reducing deployment errors by 25%.
- Contributed to responsible AI and governance initiatives, implementing bias detection, fairness testing, and model explainability reports aligned with Microsoft's AI principles for ethical deployment.

TECHNICAL SKILLS

Programming & Frameworks: Python (FastAPI, Flask), R, SQL, Java, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, OpenCV, spaCy, Transformers (Hugging Face), NumPy, Pandas, Matplotlib, Seaborn

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning (CNN, RNN, Transformer Architectures), Feature Engineering, Model Evaluation & Optimization, Computer Vision, NLP (NER, Summarization, Sentiment Analysis), Recommendation Systems, Hyperparameter Tuning

Generative AI & LLMs: LLMs (GPT, Llama, Claude, Gemini, Mixtral, Qwen), LangChain, LangGraph, LlamaIndex, NVIDIA NeMo, RAG Pipelines, Multi-Modal AI (Text, Image, Tabular), Fine-tuning (LoRA, QLoRA), RLHF, Constitutional AI, Prompt Engineering, CrewAI, Vector Databases (FAISS, Pinecone, Milvus), Retrieval Fusion, Safety & Red-Teaming

MLOps & Model Deployment: MLflow, Kubeflow, Airflow, DVC, CI/CD (GitHub Actions, Azure DevOps), Docker, Kubernetes (EKS, OpenShift), Helm, ArgoCD, Git, GitOps, Terraform, Ansible, Model Registry, Canary Deployments, Automated Retraining Pipelines

Performance & GPU Optimization: NVIDIA Triton Inference Server, TensorRT, DeepSpeed, bitsandbytes, CUDA, VLLM, Quantization (INT8/FP16), Speculative Decoding, Distributed Training, Parallel Inference

Cloud & Big Data Platforms: AWS (S3, EC2, Lambda, SageMaker, Bedrock), Azure (ML, Cognitive Services, Data Factory, Databricks), GCP (Vertex AI, BigQuery, Dataflow), Snowflake, Delta Lake, Apache Spark, Flink, Kafka, NATS JetStream, gRPC, REST APIs, Event-Driven Architectures

Model Monitoring & Responsible AI: SHAP, LIME, Captum, Drift Detection, Fairness & Bias Audits, Explainability Dashboards, Responsible AI Frameworks, Model Transparency, Observability (Prometheus, Grafana, NVIDIA Morpheus)

Data Engineering & Streaming: ETL/ELT Pipelines, Data Preprocessing, Feature Stores, Real-time Ingestion, Batch & Stream Processing, Data Normalization, Redis, Message Queues, Data Validation Frameworks

Visualization & Reporting: Tableau, Power BI, Looker, Plotly, Jupyter, Interactive ML Dashboards, Metric Tracking with W&B & LangSmith

Methodologies & Tools: Agile/Scrum, Cross-functional Collaboration, AI Governance, Data Privacy (GDPR, CCPA), A/B Testing, Experiment Tracking, Documentation Automation

EDUCATION

Master of Science in Computer Science | GPA: 4.0/4.0
Texas A&M University-Kingsville | Kingsville, TX

CERTIFICATES

AWS Certified Solutions Architect - Associate
Career Essentials in Generative AI - Microsoft and LinkedIn

PROJECTS

Enterprise Knowledge Graph RAG System | Intelligent Knowledge Assistant
Tech Stack: LangGraph, Neo4j, Pinecone, GPT-4, Claude-3.5-Sonnet, NVIDIA NeMo, FastAPI, AWS Bedrock
Designed an enterprise-grade knowledge assistant using LangGraph and Neo4j for context-aware document retrieval, improving enterprise knowledge accessibility and reducing manual research time by 55%.
Integrated hybrid dense-sparse retrieval via Pinecone and Cohere re-ranking to enhance contextual accuracy and citation relevance in multi-turn question answering.
Implemented multi-agent collaboration with specialized retrieval, summarization, and validation agents using LangGraph, enabling autonomous research synthesis and fact verification.
Deployed scalable inference APIs on AWS Bedrock and FastAPI with role-based access control, supporting 10K+ concurrent user queries across departments.

Generative AI Model Alignment Platform | RLHF-Driven LLM Evaluation & Fine-Tuning
Tech Stack: PyTorch, Hugging Face, RLHF, LoRA/QLoRA, VLLM, Constitutional AI, TensorRT-LLM, MLflow
Engineered an RLHF-based alignment platform for fine-tuning open-weight LLMs (Llama-3, Mixtral) using preference datasets and constitutional constraints, improving factual consistency by 37%.
Automated model safety evaluation via adversarial red-teaming, bias detection, and preference scoring, achieving 25% fewer hallucinations in production outputs.
Deployed optimized inference with TensorRT-LLM and VLLM speculative decoding, reducing latency by 40% while maintaining model accuracy under real-time workloads.
Integrated MLflow and W&B for experiment tracking, model evaluation, and governance reporting, ensuring traceability and reproducibility across fine-tuning workflows.